COURSE DESCRIPTION			
Program	**Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Civil Engineering (Construction Technology)**		
Course Title	**Construction Materials Lab**		
Course Code	**2018**	**Semester**	**II**
Type of Course	**Lab**	**Contact Hours**	**45**
Teaching Scheme (L: T:P:R)	**0:0:3:0**	**Credits**	**1.5**
CIE Marks	**60**	**ESE Marks**	**40**

Introduction

Construction Materials Lab is a practical course that provides hands-on experience with commonly used construction materials and basic construction activities. It helps students apply theoretical knowledge to real site practices and develop essential skills required in the construction industry.

Course Objectives

1	To provide hands-on training in basic construction activities.
2	To enable students to identify and use natural, artificial, and processed construction materials effectively.
3	To develop practical skills and workmanship by applying theoretical knowledge to real construction practices.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Building Construction	1011	Fundamentals of Civil Engineering	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Perform basic field and laboratory tests on bricks and execute setting out of a simple building.	Apply
CO2	Identify various types of soils and flooring tiles.	Apply
CO3	Identify special processed construction materials and evaluate sand quality through various tests.	Apply
CO4	Determine the bulk density of aggregates and perform bar bending.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	3	3	-	-	-	-	-	-
CO2	3	-	-	3	3	-	-	-	-	-	-
CO3	3	-	-	3	3	-	-	-	-	-	-
CO4	3	-	-	3	3	-	-	-	-	-	-
Total	12	-	-	12	12	-	-	-	-	-	-
Correlation Level	3	-	-	3	3	-	-	-	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Determination of dimension, weight and density of bricks.	Measure dimensions of 10 bricks and find average dimension, weight & density.	CO1	Apply	3
2	Field test on bricks	Perform field tests - dropping, striking and scratching by nail and correlate the results obtained.	CO1	Apply	3
3	Cement Mortar	Prepare the cement mortar of proportion 1:3 or 1:6 using cement and sand.	CO1	Apply	3
4	Setting out of building	Setting out a small building in the field.	CO1	Apply	3
Module II					
5	Report preparation on types of soil in foundation pit	Identify various layers and types of soil in the foundation pit and prepare a report.	CO2	Apply	3
6	Identification & Report preparation on flooring tiles	Identify different types of flooring tiles such as vitrified tiles, ceramic tiles, glazed tiles, mosaic tiles, paving blocks and prepare report about the specifications.	CO2	Apply	3
	Series Exam - I		CO2	Remember	
Module III					
7	Special processed construction materials	Identify and make a report on different types of special processed construction materials relevant for building construction and compare their features.	CO3	Apply	3
8	Bulking	Bulking of Sand	CO3	Apply	3
9	Silt Content	Determine Silt Content in Sand	CO3	Apply	3
Module IV					
10	Bulk Density - Fine	Determine Bulk density of fine aggregate	CO4	Apply	3
11	Bulk Density - Coarse	Determine Bulk density of coarse aggregate	CO4	Apply	3
12	Bar bending	Perform bar bending for Hooks, Stirrups, Ties and Crank bars	CO4	Apply	3
	Open Ended Experiment		CO4	Apply	3
	Series Exam - II		CO4	Apply	3

Open-ended experiment

Objective:
To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:
At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:
Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Interpreting the theory concepts and finishing lab record.	1.5
2	Interpreting the theory concepts and finishing lab record.	1.5
3	Refer IS codes & understand the physical properties and specifications of cement used in the preparation of cement mortar	1.5
4	Identify and list out different methods of setting out of buildings.	1.5
Module II		

5	Interpreting the theory concepts and finishing lab record.	1.5
6	Interpreting the theory concepts and finishing lab record.	1.5
7	Study the theory concepts and prepare for examination	1.5
Module III		
8	Interpreting the theory concepts and finishing lab record.	1.5
9	Calculation, interpretation of results and finishing lab record.	1.5
10	Calculation, interpretation of results and finishing lab record.	1.5
Module IV		
11	Calculation, interpretation of results and finishing lab record.	1.5
12	Calculation, interpretation of results and finishing lab record.	1.5
13	Interpreting the theory concepts and finishing lab record.	1.5
14	Study the theory concepts and prepare for examination	1.5
15	Identification and study of engineering problem and report preparation.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Construction Materials	Ghose, D. N	Tata McGraw Hill, New Delhi.
2	Civil Engineering Construction Materials	S.K. Sharma	Khanna Publishing House, New Delhi

REFERENCE			
Sl. No.	Title	Author	Publication
1	Building Materials	Varghese, P.C.	PHI learning, New Delhi
2	Engineering Material	Rangwala, S.C	Charator publisher, Ahemdabad.
3	Building Materials	Duggal, S. K	New International, New Delhi.

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://wecivilengineers.wordpress.com/2018/03/28/how-to-calculate-quantity-of-cement-sand-water-in-mortar-of-13/	I
2	https://youtu.be/aEMMAT5E_Ow?si=_s1u8DhgTpaRz2MK	I
3	https://gharpedia.com/blog/types-of-soil-for-building-foundation/	II
4	https://wecivilengineers.wordpress.com/2018/10/15/bulking-of-sand-test/	III
5	https://thecivilengineerings.com/bar-bending-schedule-formulas-bbs-formula/	IV
6	https://bis.gov.in (IS Codes)	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Weighing balance	Digital, least count 1 g	3
2	Measuring tape	30 m	1
3	Mixing trays, Measuring jars, trowels, tamping rods , bar bending tool	Standard	Adequate

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student’s ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50